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(54) **CABLE MANAGEMENT DEVICE**

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H02G 3/263 (2013.01); **H02G 3/30** (2013.01)

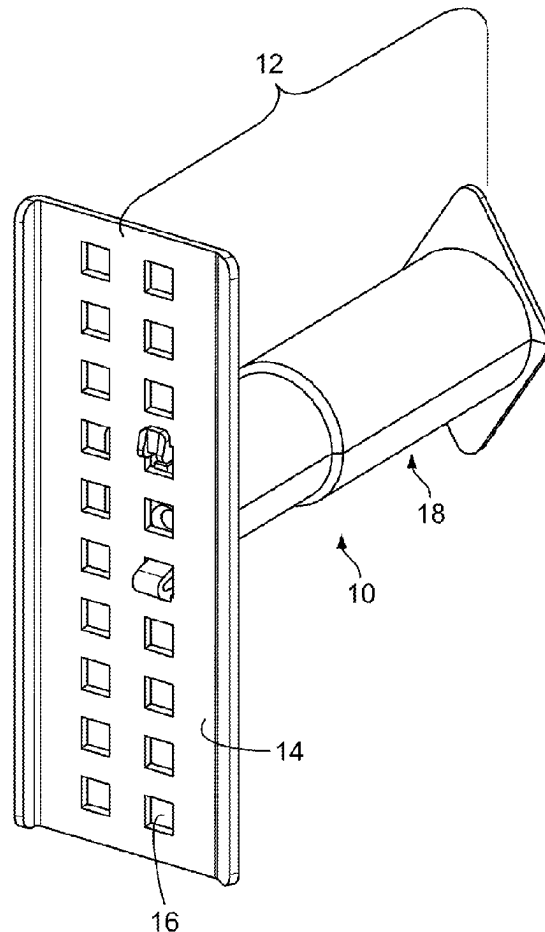
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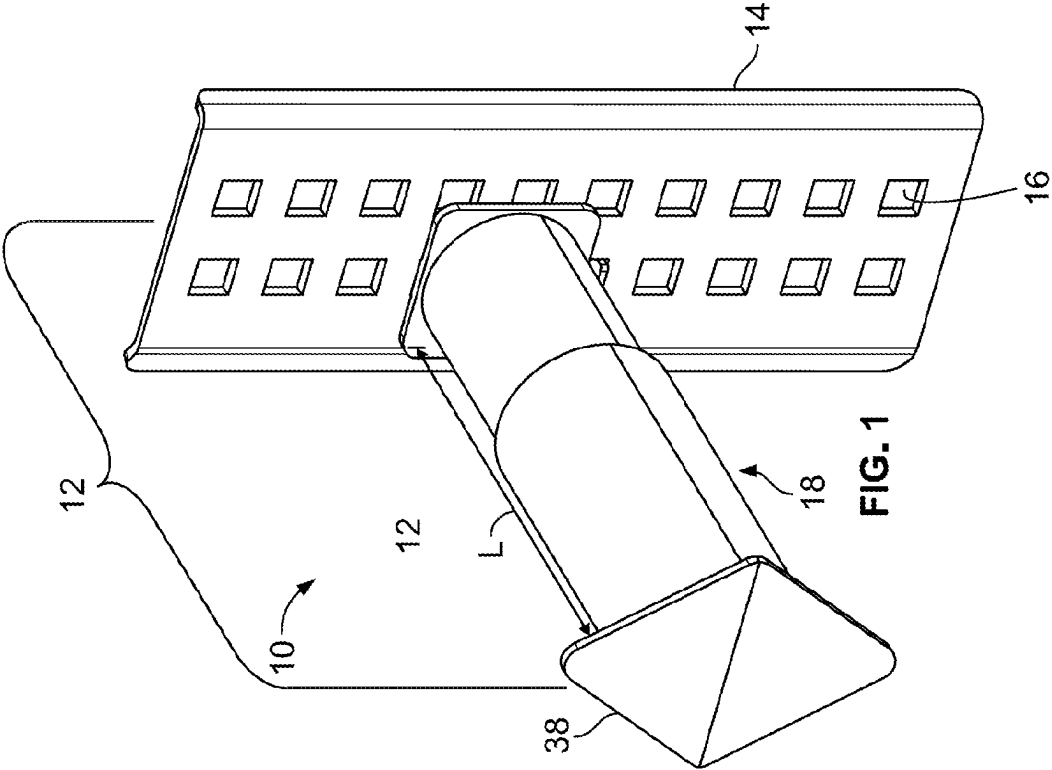
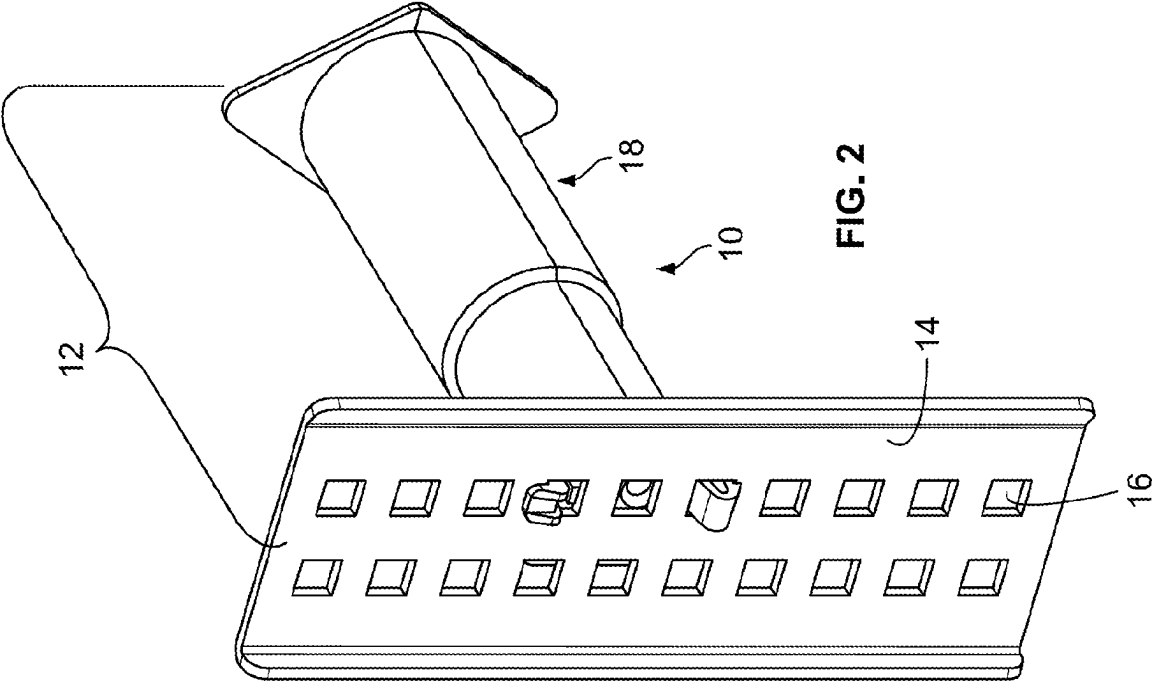
Related U.S. Application Data

(63) Continuation of application No. 18/668,541, filed on May 20, 2024, now Pat. No. 12,298,582, which is a continuation of application No. 18/323,673, filed on May 25, 2023, now Pat. No. 11,988,884, which is a continuation of application No. 17/586,518, filed on Jan. 27, 2022, now Pat. No. 11,668,888, which is a continuation of application No. 16/902,770, filed on Jun. 16, 2020, now Pat. No. 11,262,518, which is a continuation of application No. 16/324,191, filed on Feb. 8, 2019, filed as application No. PCT/US2017/045703 on Aug. 7, 2017, now Pat. No. 10,690,874.

(57) **ABSTRACT**

A cable management device for mounting to a telecommunications fixture includes an outer barrel disposed over an inner barrel, one of the outer barrel and the inner barrel defining a plurality of discrete detents positioned in a stacked arrangement axially along an length thereof, and the other of the outer barrel and the inner barrel defining at least one flexible cantilever arm defining a tab configured to lock into a selected one of the detents for allowing adjustment of a length of the cable management device.





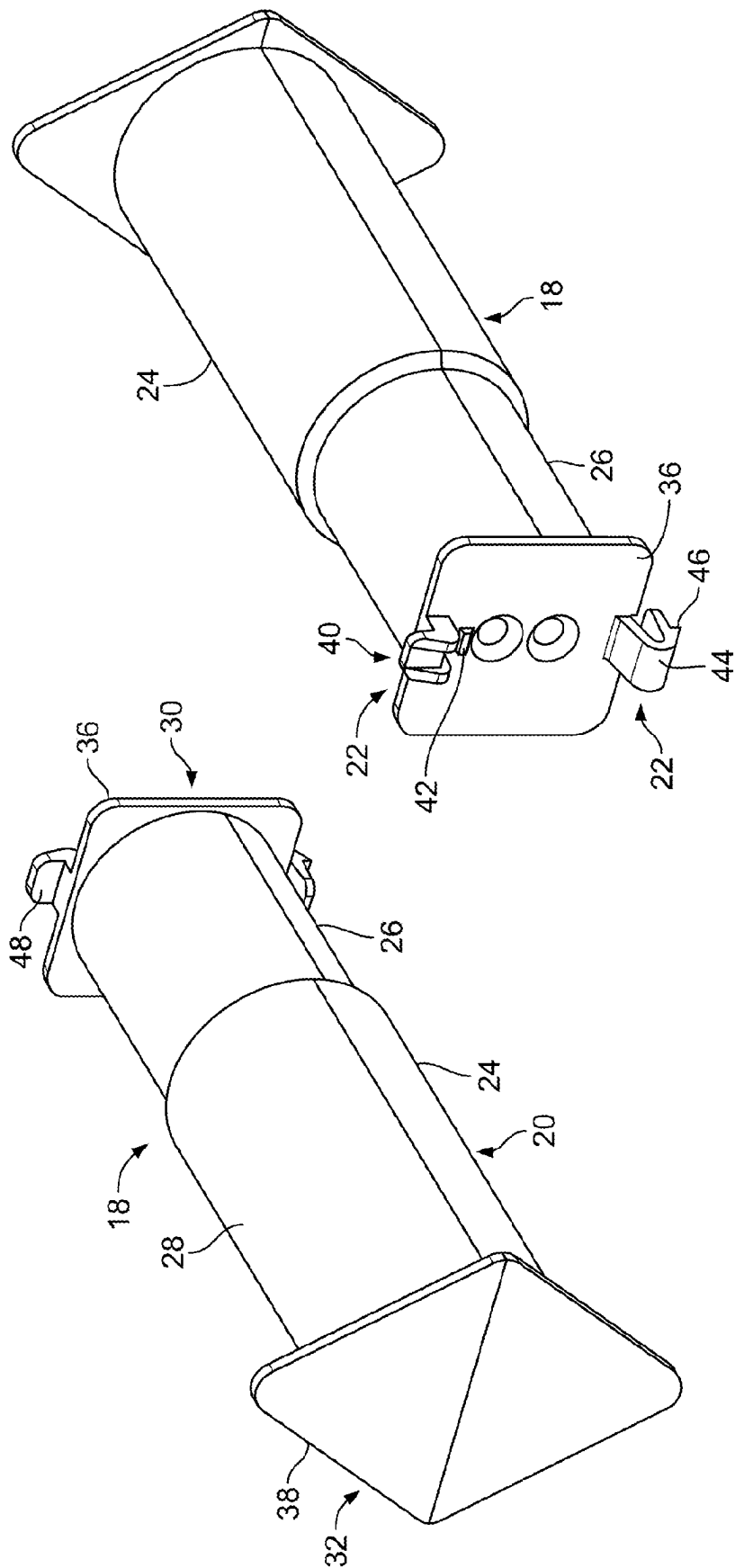


FIG. 4

FIG. 3

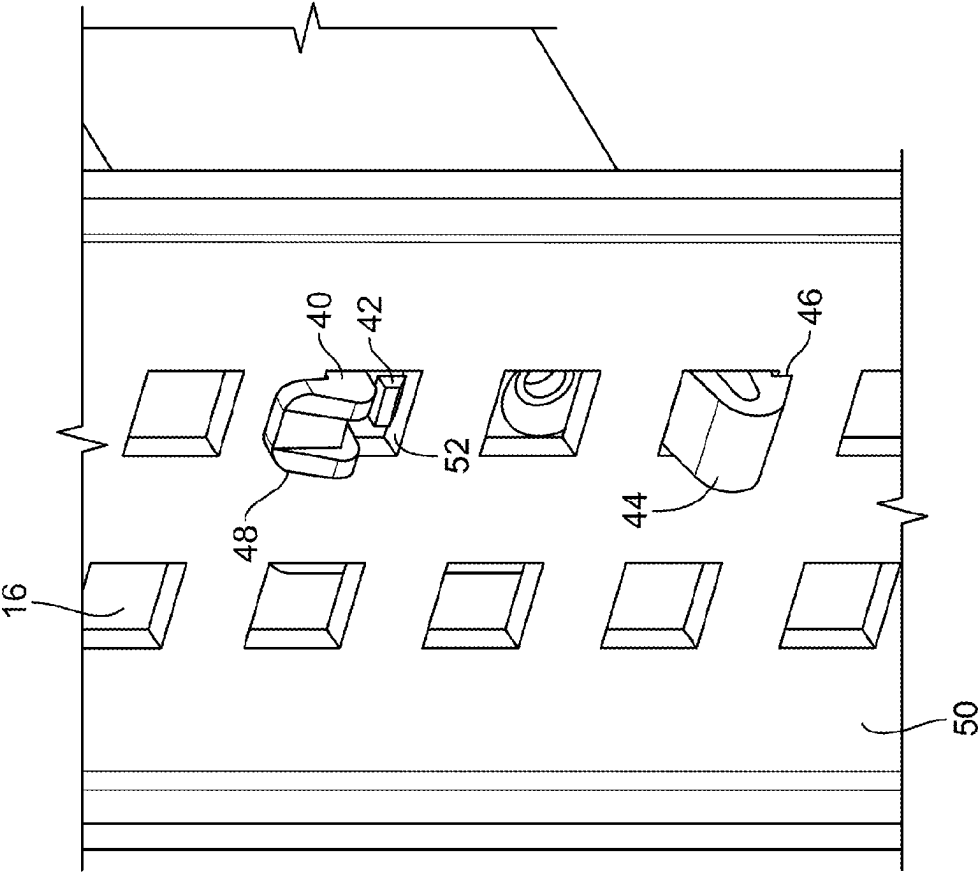


FIG. 6

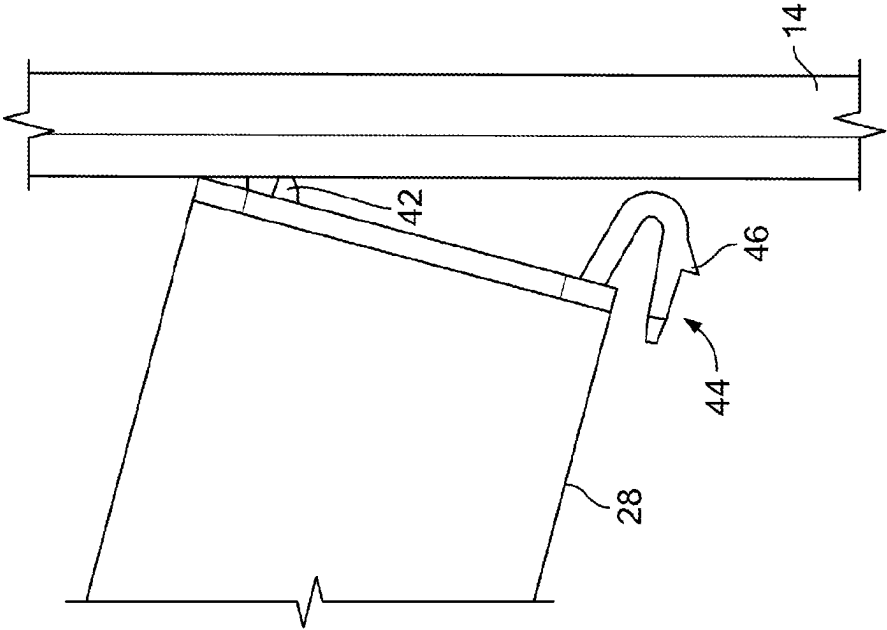


FIG. 5

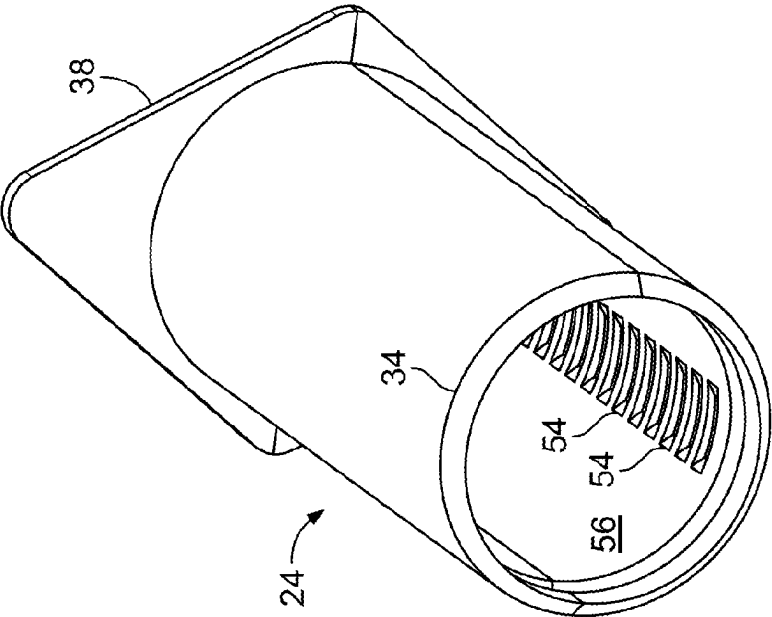


FIG. 7

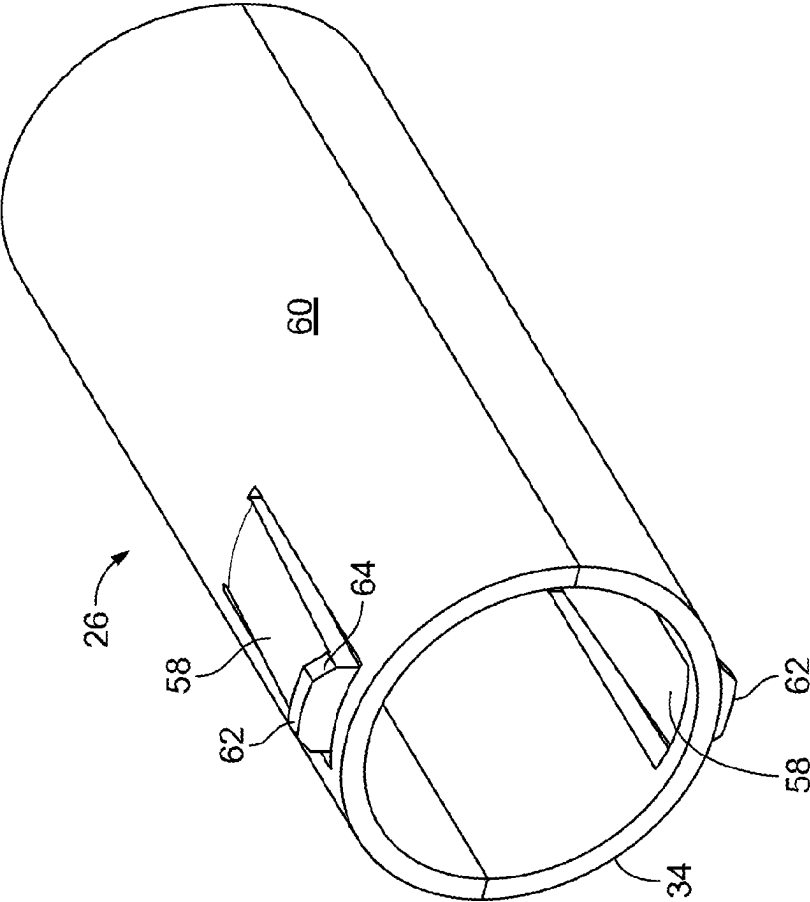


FIG. 8

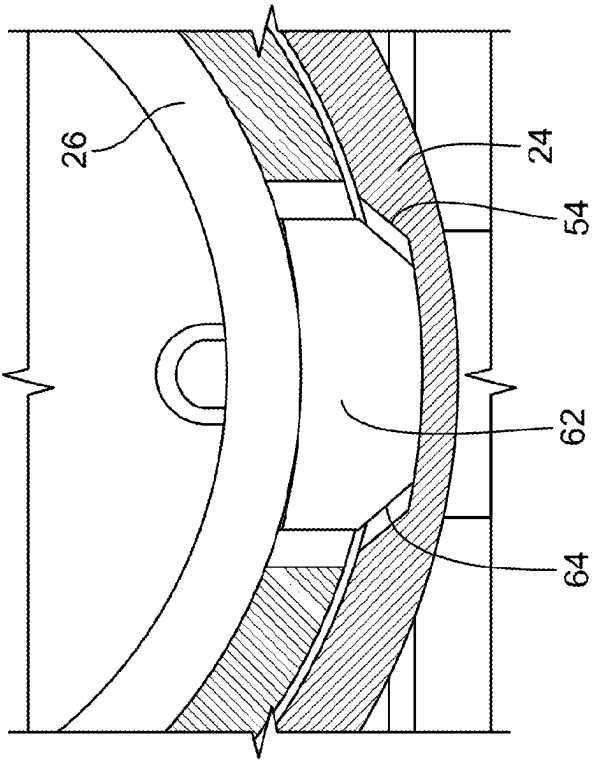


FIG. 10

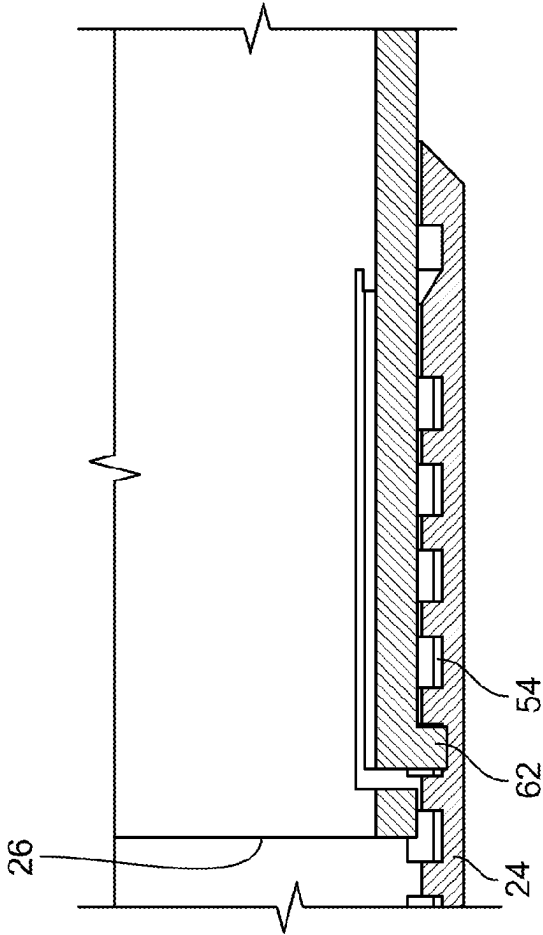


FIG. 9

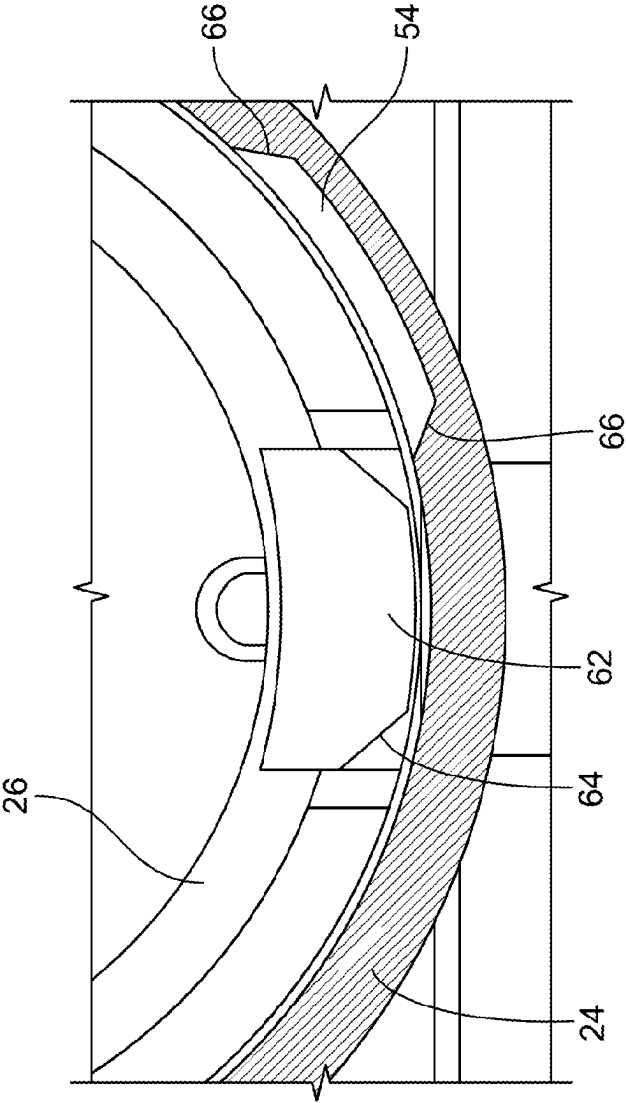
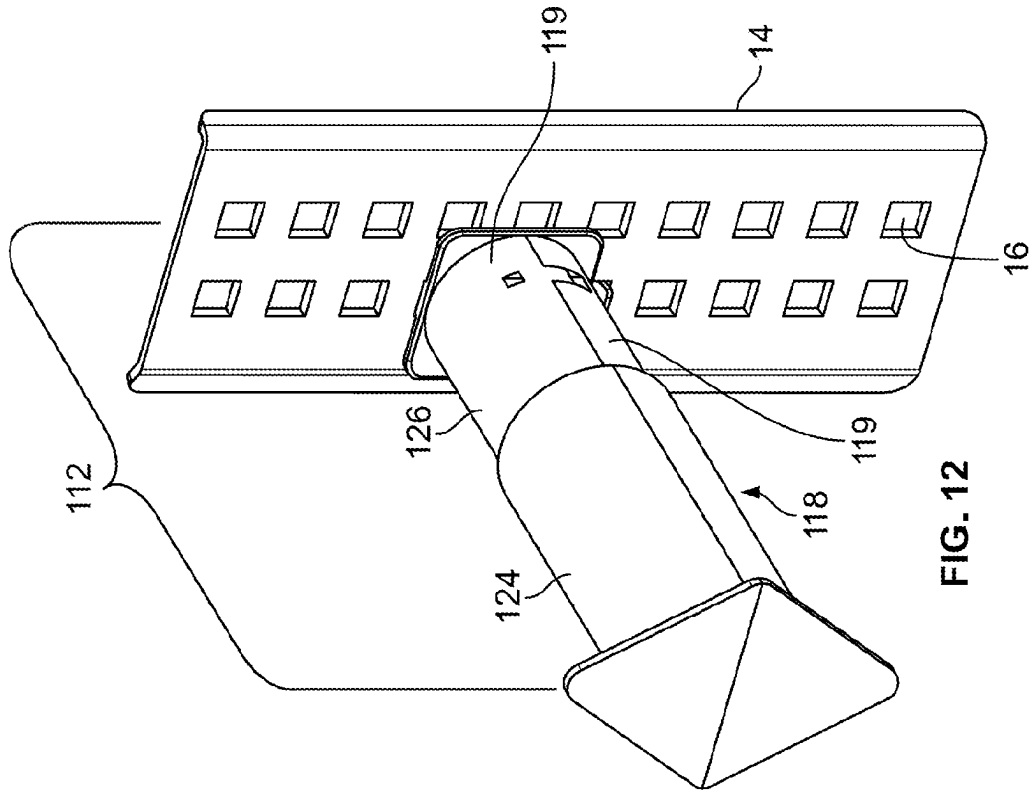
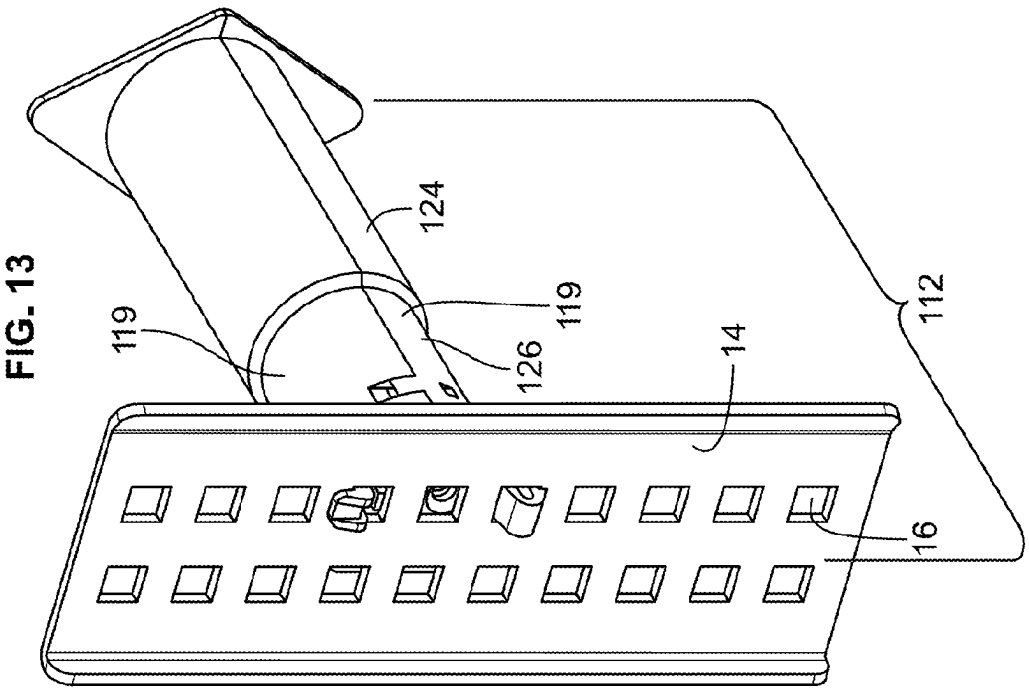


FIG. 11



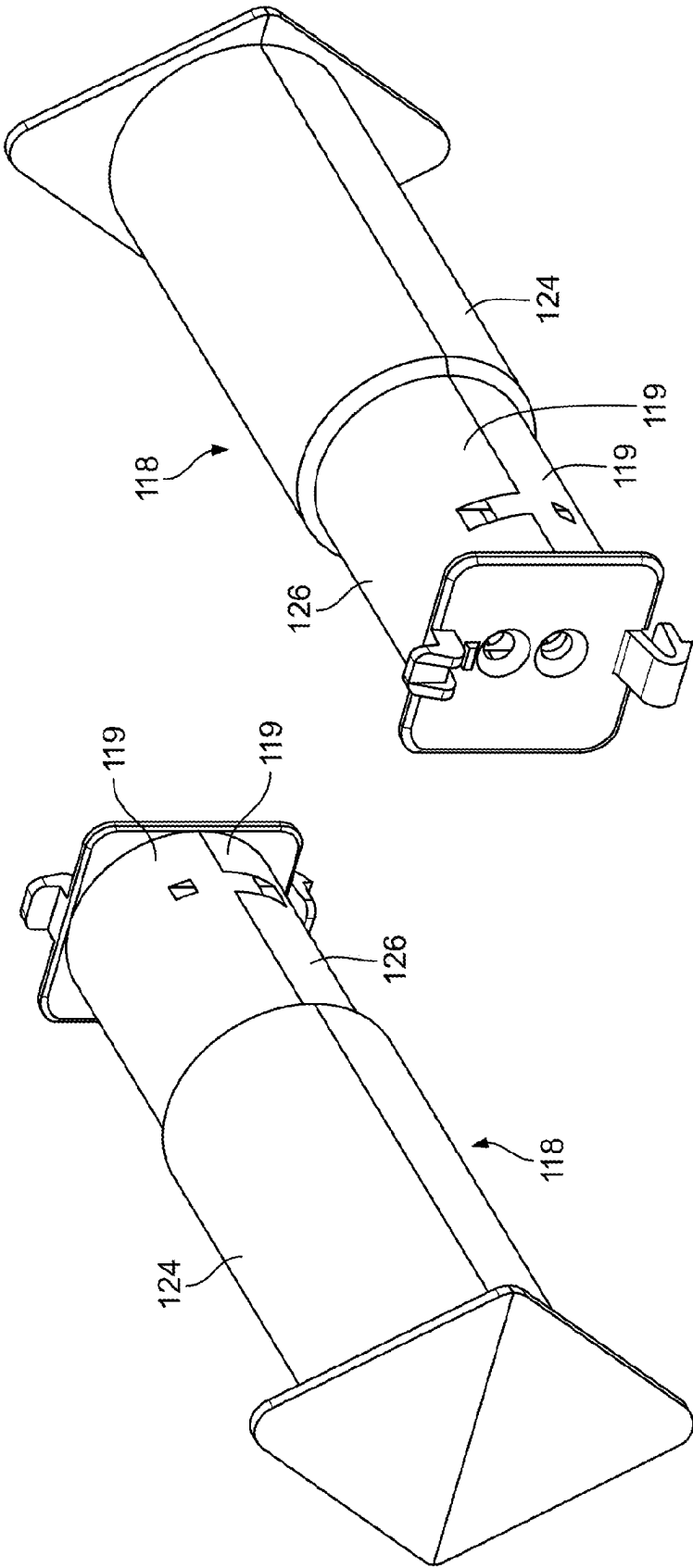


FIG. 15

FIG. 14

CABLE MANAGEMENT DEVICE**CROSS-REFERENCE TO RELATED APPLICATION**

[0001] This application is a continuation of U.S. patent application Ser. No. 18/668,541, filed on May 20, 2024; which is a continuation of U.S. patent application Ser. No. 18/323,673, filed on May 25, 2023, now U.S. Pat. No. 11,988,884; which is a continuation of U.S. patent application Ser. No. 17/586,518, filed on Jan. 27, 2022, now U.S. Pat. No. 11,668,888; which is a continuation of U.S. patent application Ser. No. 16/902,770, filed on Jun. 16, 2020, now U.S. Pat. No. 11,262,518; which is a continuation of U.S. patent application Ser. No. 16/324,191, filed on Feb. 8, 2019, now U.S. Pat. No. 10,690,874; which is a National Stage Application of PCT/US2017/045703, filed on Aug. 7, 2017; which claims the benefit of U.S. Patent Application Ser. No. 62/372,123, filed on Aug. 8, 2016, the disclosures of which are incorporated herein by reference in their entireties. To the extent appropriate, a claim of priority is made to each of the above disclosed applications.

TECHNICAL FIELD

[0002] The present disclosure relates generally to devices for enhancing cable management of a telecommunications system. More particularly, the present disclosure relates to a cable management assembly including a cable management spool used on a telecommunications panel.

BACKGROUND

[0003] Telecommunications systems commonly include cables (e.g., fiber optic cables, or copper twisted pair cables) for interconnecting pieces of telecommunications equipment. Telecommunications racks typically hold a variety of different pieces of telecommunications equipment. Often thousands of cables are used to interconnect the various pieces of telecommunications equipment mounted on the racks.

[0004] Because of the large number of cables associated with telecommunications equipment, cable management is crucial. Cable management involves efficiently routing cables to minimize the space occupied by such cables. Cable management also involves routing cables in an orderly manner so as to reduce the likelihood of cable tangling. Ease of cable organization is also a factor related to effective cable management.

[0005] Cable management is also important in preventing damage to the cables. Unnecessary or excessive displacement of fiber optic cables, for example, is undesirable. As the fiber optic cables are displaced, they are subject to bending and other forces. Bending of the fibers can cause attenuation and loss of signal strength. As the fiber bends, the fiber can also break, resulting in a loss of transmission through the fiber.

[0006] According to one example type of cable management, cable spools may extend from a surface in columns, rows, or other configurations. One or more fiber optic cables may be hung over or wrapped around a curved surface of one or more of these spools to take up the slack length of the fiber optic cable.

[0007] In general, cable management improvement has been sought, generally to efficiently and effectively manage

cables by providing system adaptability, and ease of cable organization, especially in such spool-based systems.

SUMMARY

[0008] One aspect of the present disclosure relates to a telecommunications system including a cable management assembly including a panel having a plurality of shaped apertures and a plurality of cable management devices in the form of spools for mounting to the panel. Each cable management device includes attachment features corresponding to the shaped apertures that secure the cable management devices to the panel at selected horizontal and vertical locations. According to one inventive aspect, the cable management devices in the form of spools may include features for adjusting or varying the axial lengths thereof for equipment clearance or space-adaptability.

[0009] According to another aspect, the disclosure relates to a cable management device in the form of a cable spool defining an outer barrel that is disposed over an inner barrel, both having a generally round configuration for providing cable management with bend radius protection. One of the outer barrel and the inner barrel includes a plurality of discretely spaced locking detents along the length thereof, and the other of the outer barrel and the inner barrel includes a flexible arm with a tab that can lock within one of the detents. Relative rotation between the outer barrel and the inner barrel allows the tab to enter or exit the detent. For example, after a desired length for the spool is determined, relative rotational movement between the outer barrel and the inner barrel allows the tab to lock within a detent that corresponds to the desired length, thus providing adjustability to the length of the overall cable management spool.

[0010] Another aspect of the present disclosure relates to a cable management assembly comprising a telecommunications panel including a plurality of shaped apertures arranged in an array of columns and rows and a plurality of cable management devices having attachment structures corresponding to the shaped apertures for removably mounting the cable management devices to the panel at selected vertical and horizontal locations, wherein each cable management device defines an outer barrel disposed over an inner barrel, one of the outer barrel and the inner barrel defining a plurality of discrete detents positioned in a stacked arrangement axially along a length thereof, and the other of the outer barrel and the inner barrel defining at least one flexible cantilever arm defining a tab configured to lock into a selected one of the detents for allowing adjustment of a length of the cable management device.

[0011] According to another aspect, the present disclosure relates to a cable management device for mounting to a telecommunications fixture, the cable management device comprising an outer barrel disposed over an inner barrel, one of the outer barrel and the inner barrel defining a plurality of discrete detents positioned in a stacked arrangement axially along a length thereof, and the other of the outer barrel and the inner barrel defining at least one flexible cantilever arm defining a tab configured to lock into a selected one of the detents for allowing adjustment of a length of the cable management device.

[0012] According to yet another aspect, the present disclosure relates to a method of using a cable management device, the method comprising adjusting a length of the cable management device by relatively sliding an outer barrel and an inner barrel of the cable management device to

select the length of the cable management device and relatively rotating the outer barrel and the inner barrel and locking a tab of a flexible cantilever arm of one of the outer barrel and the inner barrel into a detent of a plurality of discrete detents positioned axially along a length of the other of the outer barrel and the inner barrel.

[0013] A variety of inventive aspects are set forth in part in the description that follows, and in part will be apparent from the description, or may be learned by practicing various aspects of the disclosure. The aspects of the disclosure may relate to individual features as well as combinations of features. It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only, and are not restrictive of the claimed inventions.

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] FIG. 1 illustrates a telecommunications system including a cable management assembly formed from a telecommunications panel and at least one cable management device having features that are examples of inventive aspects in accordance with the present disclosure removably mounted to the panel, the cable management assembly shown from an outer perspective view.

[0015] FIG. 2 illustrates the cable management assembly of FIG. 1 from an inner perspective view.

[0016] FIG. 3 illustrates a perspective view of the cable management device of FIGS. 1-2 from a free end thereof.

[0017] FIG. 4 illustrates a perspective view of the cable management device of FIGS. 1-2 from a mounting end thereof.

[0018] FIG. 5 illustrates the mounting of the cable management device of FIGS. 1-4 to the telecommunications panel shown in FIGS. 1-2.

[0019] FIG. 6 illustrates a close-up view of the cable management device mounted to the panel of FIG. 2.

[0020] FIG. 7 illustrates a perspective view of an outer barrel of the cable management device of FIGS. 1-6.

[0021] FIG. 8 illustrates a perspective view of an inner barrel of the cable management device of FIGS. 1-6.

[0022] FIG. 9 is a partial cross-sectional view bisecting the cable management device of FIGS. 1-6 showing the outer and inner barrels thereof in a locked position.

[0023] FIG. 10 is a partial lateral cross-sectional view of the cable management device of FIGS. 1-6 showing the outer and inner barrels thereof in a locked position.

[0024] FIG. 11 is a partial cross-sectional view similar to that of FIG. 10, showing the outer and inner barrels thereof in an unlocked position.

[0025] FIG. 12 illustrates another embodiment of a telecommunications system including a cable management assembly formed from a telecommunications panel and a second example of a cable management device having features that are examples of inventive aspects in accordance with the present disclosure removably mounted to the panel, the cable management assembly shown from an outer perspective view.

[0026] FIG. 13 illustrates the cable management assembly of FIG. 12 from an inner perspective view.

[0027] FIG. 14 illustrates a perspective view of the cable management device of FIGS. 12-13 from a free end thereof.

[0028] FIG. 15 illustrates a perspective view of the cable management device of FIGS. 12-13 from a mounting end thereof.

DETAILED DESCRIPTION

[0029] Reference will now be made in detail to exemplary aspects of the present disclosure that are illustrated in the accompanying drawings. Wherever possible, the same reference numbers will be used throughout the drawings to refer to the same or like parts.

[0030] One embodiment of a telecommunications system 10 for managing cables interconnected to telecommunications components is illustrated in FIGS. 1-11. The telecommunications system 10 includes a cable management assembly 12 in accordance with the principles disclosed. The cable management assembly 12 is illustrated specifically in FIGS. 1, 2, 5, and 6 and includes a telecommunications panel 14 having a plurality of shaped apertures 16 and a plurality of cable management devices 18. In the depicted embodiment, the cable management devices 18 are provided in the form of spools 20. The panel 14 may also be referred to as a back plane or a riser.

[0031] Still referring to FIGS. 1-11, each cable management device 18 includes attachment features 22 corresponding to the shaped apertures 16 that secure the cable management devices 18 to the panel 14 at selected horizontal and vertical locations. The discrete apertures/openings 16 of the panel 14 and the cable management devices 18 are configured and sized so that a user can organize and arrange cables at various locations along the panel 14. As will be discussed in further detail, according to one inventive aspect, the cable management devices 18 may include features for adjusting or varying the axial lengths L thereof for equipment clearance or space-adaptability.

[0032] One feature of the present disclosure relates to cost reduction associated with fixed or custom-made cable management systems. The present cable management assembly 12 permits a user to configure the system 10 to a particular need. For example, in accordance with the principles disclosed herein, a user has the flexibility to position any number of the cable management devices 18 at a plurality of different locations on the panel 14 of the cable management assembly 12. In particular, the preferred panel 14 is constructed to provide a plurality of placement coordinates (i.e., vertical and horizontal locations) from which a user may choose to position the cable management devices 18. Thus, with this design, a user is not limited to a pre-made, fixed cable management system, which may not fit a particular need. Similarly, with this design, the user reduces or eliminates modification and customizing costs associated with reworking a system to fit the particular need.

[0033] Another feature of the present system 10 relates to the system's adaptability and ease of use. In one embodiment, the cable management devices 18 are detachably secured to the riser or panel 14. This feature permits a user to modify a system's cable management strategy to accommodate growth or changes needed in a quickly changing industry.

[0034] Still referring to FIGS. 1-11 in general, each cable management device 18 in the form of a spool 20 is configured to facilitate adding one or more fiber optic cables or optical fibers to a densely packed fiber storage region. The cable management device 18, in the given embodiment, is formed from an outer barrel 24 that is disposed over an inner barrel 26. The barrels 24, 26, cooperatively, define a storage surface 28 that extends between a first end 30 and a second end 32 of the spool 20.

[0035] In the example shown, both the outer barrel 24 and the inner barrel 26 forming the management device 18 define a circle-shaped lateral cross-dimensional profile 34.

[0036] At the first end 30 of the spool 20, attached to the inner barrel 26, is defined a latch plate 36. And, at the second end 32 of the spool 20, extending from the outer barrel 24, is defined a flange 38 for acting as a cable retainer to facilitate maintaining the fiber optic cables on the storage surface 28.

[0037] At the first end 30 of the spool 20, the latch plate 36 includes features 22 for latching the spool 20 to a piece of telecommunications equipment such as the panel 14 shown in FIGS. 1, 2, 5, and 6. The latch plate 36 includes a catch 40, an abutment member 42 at an upper portion thereof, and a snap-in latch member 44 at a lower portion thereof. The latch member 44 defines a latch tab 46.

[0038] The upper and the lower features 22 of the latch plate 36 are configured to cooperate with corresponding shaped apertures 16 on the panel 14 for mounting the spools 20 to desired positions on the panel 14 as discussed previously. In general, the cable management device 18 is designed to be secured to the panel 14 by insertion into the panel 14 from one side, without the use of tools or extra fastening components. As shown in detail in FIGS. 5-6, in mounting a spool 20 to a panel 14, the upper catch 40 is first inserted through an aperture 16 in the panel 14. Once a retention portion 48 of the catch 40 is hooked against the panel 14, the latch plate 36, and thus the entire spool 20, is pivoted down to insert the latch member 44 through another lower aperture 16 that is spaced apart from the upper aperture 16 of the panel 14 along the same column. The latch member 44 is flexibly inserted until the latch tab 46 clears the panel wall 50 and the spool 20 is latched into place. The abutment member 42 at the upper portion of the latch plate 36 is configured to contact a portion of the panel wall 50 defining a lower edge 52 of the upper aperture 16 and provide further stability to the latched spool 20.

[0039] In releasing the spool 20 from the panel 14 and disassembling the cable management assembly 12, the portion of the latch member 44 defining the latch tab 46 is simply flexed upwardly until the tab 46 clears the aperture 16 and the spool 20 is pivoted away from the panel 14. A final downward movement of the spool 20 in unhooking the upper catch 40 releases the spool 20 from the panel 14.

[0040] Now referring to FIGS. 7-11 specifically, as discussed above and as will be discussed in further detail below, the outer barrel 24 and the inner barrel 26 include features that provide adjustability of the length L of the overall spool 20. Allowing adjustability of the length L of the spool 20 facilitates clearance for other equipment that may be positioned adjacent the panel 14.

[0041] In the example shown, the outer barrel 24 defines a plurality of discretely positioned detents 54 at the inner surface 56 thereof extending axially along the outer barrel 24. It should be noted that two sets of detents 54 may be positioned at the inner surface 56, on opposing sides thereof.

[0042] In the example shown, the inner barrel 26 defines a flexible cantilever arm 58 on opposing sides of the outer surface 60 thereof. The flexible cantilever arms 58 define tabs 62 that are configured to project radially outwardly from the outer surface 60 of the inner barrel 26. As will be discussed, the tabs 62 are configured to mate with and lock into the detents 54 provided on the inner surface 56 of the outer barrel 24 in positionally locking the outer barrel 24 to

the inner barrel 26. When the tabs 62 are not within the detents 54, the outer barrel 24 can be slidably moved in a telescoping fashion with respect to the inner barrel 26 in adjusting the length L of the spool 20. And, once the desired length L is set and the tabs 62 are in alignment with the detents 54 that correspond to the selected length L of the spool 20, relative rotation of the outer barrel 24 over the inner barrel 26 moves the tabs 62 into the detents 54 to lock the outer barrel 24 with respect to the inner barrel 26.

[0043] If further adjustment is needed, relative rotation between the outer barrel 24 and the inner barrel 26 once again allows a tab 62 or a pair of tabs 62 to exit a detent 54 or a pair of detents 54. After a desired length L for the spool 20 is determined, relative rotational movement between the outer barrel 24 and the inner barrel 26 once again allows the tab 62 to lock within another of the detents 54 that corresponds to the desired length L. Thus, the telescoping slidable movement and the rotational locking movement between the outer and inner barrels 24, 26 provide adjustability to the length L of the overall cable management spool 20.

[0044] FIGS. 9-10 illustrate a locked position for the spool 20 with one of the tabs 62 of the flexible cantilever arms 58 within one of the detents 54. FIG. 11 illustrates an unlocked position for the spool 20, wherein a tab 62 or a pair of tabs 62 are outside of a given detent 54 or pair of detents 54. In the unlocked position shown in FIG. 11, the outer barrel 24 is able to telescopically slide with respect to the inner barrel 26 so as to provide adjustment for the length L of the spool 20.

[0045] It should be noted that the tabs 62 on the flexible cantilever arms 58 may include tapered surfaces 64 to help with slidable insertion into and removal from the detents 54. As shown in the cross-sectional views in FIGS. 10 and 11, the detents 54 may also define tapered entry/exit surfaces 66 to facilitate insertion and removal of the tabs 62.

[0046] It will be appreciated that, even though in the depicted embodiment, the outer barrel 124 is the structure that is configured with the detents 54 and the inner barrel 26 is the structure that is configured with the cooperating cantilever arms 58, in other embodiments, these structures may be reversed.

[0047] FIGS. 12-15 illustrate a cable management assembly 112 including a panel 14 and a cable management device 118 that has features similar to those described above with respect to the device 18 of FIGS. 1-11. In the device 118 of FIGS. 12-15, both the outer and the inner barrels 124, 126 of the device 118 are depicted as having been molded from two semi-circular pieces 119 that are fitted together with a snap-fit interlock to form the circular lateral cross-dimensional profile 34. In the embodiment of the spool 20 illustrated in FIGS. 1-11, the outer and the inner barrels 24, 26 are illustrated as structures that are molded as a single piece. Otherwise, the devices 18, 118 of FIGS. 1-11 and 12-15 function in a similar manner.

[0048] The above specification provides a complete description of the telecommunications systems including the cable management assemblies and methods of use. Since many embodiments of the disclosure can be made without departing from the spirit and scope of the inventive aspects, the inventive aspects reside in the claims hereinafter appended.

1-20. (canceled)

21. A cable management assembly comprising:

a telecommunications frame defining a plurality of apertures; and

a plurality of cable management devices mounted to the telecommunications frame, the cable management devices having attachment structures corresponding to the apertures defined by the telecommunications frame for removably mounting the cable management devices to the telecommunications frame at selected locations on the telecommunications frame, wherein each cable management device defines a first cylindrical structure defining a wall with a circular lateral cross-dimensional profile, wherein each cable management device also defines a second structure defining a wall with a circular lateral cross-dimensional profile, the second structure positioned at least partially within the first cylindrical structure, the first cylindrical structure defining at least one detent that is cut at least partially into the wall of the first cylindrical structure and the second structure defining at least one tab protruding radially outwardly from the wall of the second structure that is configured to lock into the at least one detent of the first cylindrical structure to prevent linear movement of the second structure relative to the first cylindrical structure, wherein the second structure is configured to be rotatable relative to the first cylindrical structure for disengaging the at least one tab from the at least one detent for allowing linear movement of the second structure relative to the first cylindrical structure for adjusting an overall length of the cable management device.

22. The cable management assembly of claim 21, wherein the first cylindrical structure defines a plurality of discrete detents that are cut at least partially into the wall of the first cylindrical structure, wherein the at least one tab of the second structure is configured to lock into a selected one of the detents for allowing adjustment of the overall length of the cable management device.

23. The cable management assembly of claim 21, wherein the apertures of the telecommunications frame and the attachment structures of the cable management devices are configured such that the cable management devices can be removably mounted to the telecommunications frame at selected vertical and horizontal locations on the frame.

24. The cable management assembly of claim 21, wherein the second structure is slidable with respect to the first cylindrical structure when the at least one tab is out of the at least one detent for adjusting the length of the cable management device.

25. The cable management assembly of claim 21, wherein the first cylindrical structure defines two sets of detents cut at least partially into opposing sides of the wall of the first cylindrical structure, and the second structure defines two tabs that are protruding from opposing sides of the wall of the second structure, wherein the tabs of the second structure are configured to lock into a selected one of the detent sets for allowing adjustment of the overall length of the cable management device.

26. The cable management assembly of claim 21, wherein one of the first cylindrical structure and the second structure includes a cable retention flange at an end of the cable management device.

27. The cable management assembly of claim 21, wherein the at least one tab of the second structure is locked into the at least one detent of the first cylindrical structure under a bias force.

28. The cable management assembly of claim 27, wherein the at least one tab of the second structure is locked into the at least one detent of the first cylindrical structure under a bias force that is along a radial direction relative to the walls of the first cylindrical structure and the second structure.

29. A cable management device for mounting to a telecommunications fixture, the cable management device comprising:

a first cylindrical structure defining a wall with a circular lateral cross-dimensional profile; and

a second structure defining a wall with a circular lateral cross-dimensional profile, the second structure positioned at least partially within the first cylindrical structure;

wherein the first cylindrical structure defines at least one detent that is cut at least partially into the wall of the first cylindrical structure and the second structure defines at least one tab protruding radially outwardly from the wall of the second structure that is configured to lock into the at least one detent of the first cylindrical structure to prevent linear movement of the second structure relative to the first cylindrical structure, wherein the second structure is configured to be rotatable relative to the first cylindrical structure for disengaging the at least one tab from the at least one detent for allowing linear movement of the second structure relative to the first cylindrical structure for adjusting an overall length of the cable management device.

30. The cable management device of claim 29, wherein the first cylindrical structure defines a plurality of discrete detents that are cut at least partially into the wall of the first cylindrical structure, wherein the at least one tab of the second structure is configured to lock into a selected one of the detents for allowing adjustment of the overall length of the cable management device.

31. The cable management device of claim 29, wherein the second structure is slidable with respect to the first cylindrical structure when the at least one tab is out of the at least one detent for adjusting the length of the cable management device.

32. The cable management device of claim 29, wherein the first cylindrical structure defines two sets of detents cut at least partially into opposing sides of the wall of the first cylindrical structure, and the second structure defines two tabs that are protruding from opposing sides of the wall of the second structure, wherein the tabs of the second structure are configured to lock into a selected one of the detent sets for allowing adjustment of the overall length of the cable management device.

33. The cable management device of claim 29, wherein one of the first cylindrical structure and the second structure includes a cable retention flange at an end of the cable management device.

34. The cable management device of claim 29, wherein the at least one tab of the second structure is locked into the at least one detent of the first cylindrical structure under a bias force.

35. The cable management device of claim 34, wherein the at least one tab of the second structure is locked into the at least one detent of the first cylindrical structure under a

bias force that is along a radial direction relative to the walls of the first cylindrical structure and the second structure.

36. A cable management device for mounting to a telecommunications fixture, the cable management device comprising:

- a first cylindrical structure defining a wall with a circular lateral cross-dimensional profile; and
- a second structure defining a wall with a circular lateral cross-dimensional profile, the second structure positioned at least partially within the first cylindrical structure;

wherein one of the first cylindrical structure and the second structure defines a plurality of discrete detents that are cut at least partially into the wall of one of the first cylindrical structure and the second structure, wherein the other of the first cylindrical structure and the second structure defines at least one tab protruding radially from the wall of the other of the first cylindrical structure and the second structure that is configured to lock into a selected one of the discrete detents to prevent linear movement of the second structure relative to the first cylindrical structure, wherein the at least one tab is configured to be locked into a selected one of the discrete detents under a bias force, wherein the second structure is configured to be rotatable relative to the first cylindrical structure for disengaging the at least one tab from the selected detent for allowing linear movement of the second structure relative to the first cylindrical structure for adjusting an overall length of the cable management device.

37. The cable management device of claim **36**, wherein the second structure is slidable with respect to the first cylindrical structure when the at least one tab is out of the selected detent for adjusting the length of the cable management device.

38. The cable management device of claim **36**, wherein the first cylindrical structure defines the plurality of discrete detents cut at least partially into the wall of the first cylindrical structure, and the second structure defines the at least one tab protruding radially from the wall of the second structure.

39. The cable management device of claim **38**, wherein the first cylindrical structure defines two sets of detents cut at least partially into opposing sides of the wall of the first cylindrical structure, and the second structure defines two tabs that are protruding from opposing sides of the wall of the second structure, wherein the tabs of the second structure are configured to lock into a selected one of the detent sets for allowing adjustment of the overall length of the cable management device.

40. The cable management device of claim **36**, wherein one of the first cylindrical structure and the second structure includes a cable retention flange at an end of the cable management device.

41. The cable management device of claim **36**, wherein the at least one tab is configured to be locked into a selected one of the discrete detents under a bias force that is along a radial direction relative to the walls of the first cylindrical structure and the second structure.

* * * * *